GRADIENT-05

GRADIENT

HIGH-CONTROL 6-AXIS ARM

GRADIENTOS INTEGRATED

KEY BENEFITS

- Six-axis articulated architecture for flexible tool positioning.
- EtherCAT RTCore execution for coordinated servo control.
- Numeric QuIK IK backend for Cartesian motion planning.
- Direct joint-to-actuator mapping for transparent diagnostics.
- 17-bit motor-side absolute encoder feedback model.

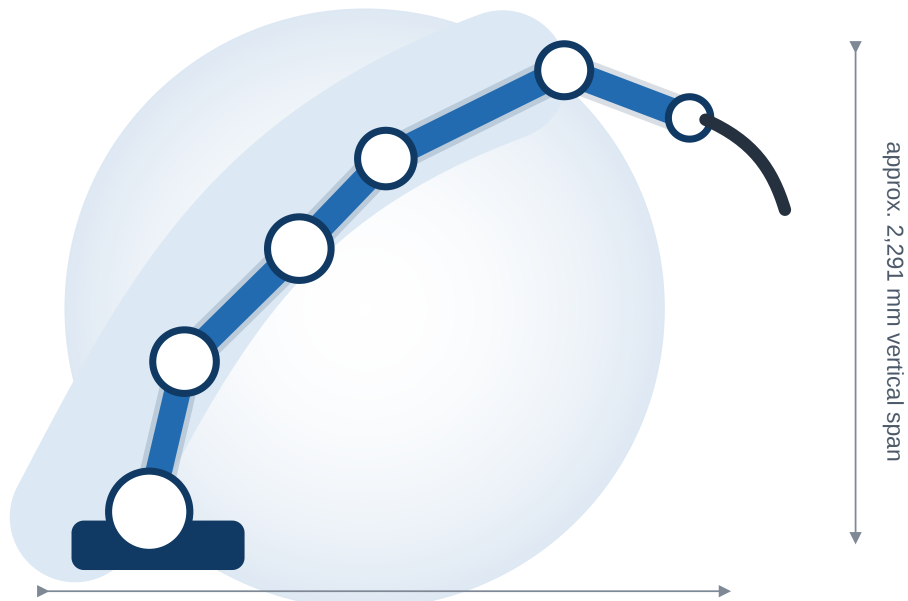
SPECIFICATIONS

Axes	6
Reach	approx. 1,509 mm
Vertical span	approx. 2,291 mm
Encoder	131,072 counts/rev
Payload	**In validation**
Repeatability	**In validation**

APPLICATION

Motion-control development, robotic process automation, end-effector integration, and advanced robotic workflow R&D.

Gradient-05 reach envelope



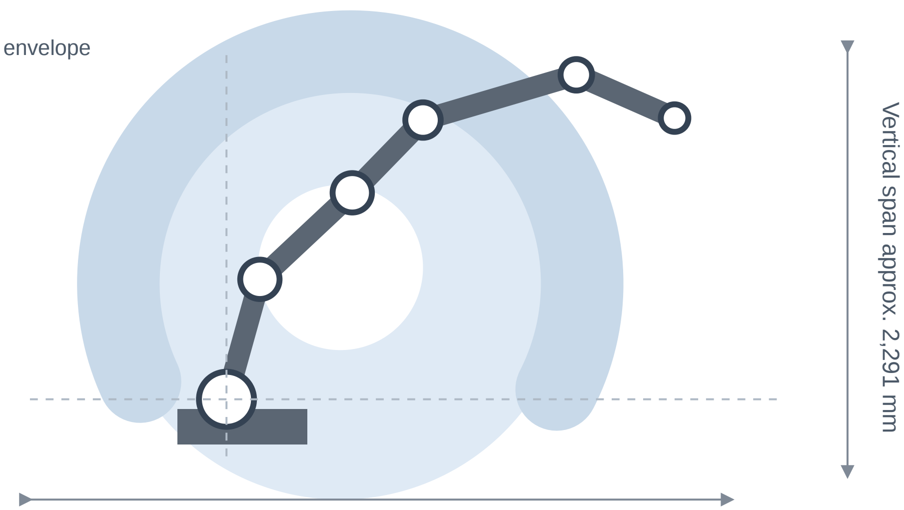
- Extended rotation ranges support base and wrist workflows beyond one revolution.
- Integrated commissioning exposes native-home setup, drive status, and actuator diagnostics.
- Current GradientOS kinematics define a broad development envelope for process and tooling work.
- Controller defaults include 0.1 m/s profile velocity and 0.05 m/s^2 profile acceleration.
- Cartesian jog caps are 0.2 m/s linear and 180 deg/s angular.
- Payload, repeatability, mass, flange, power, and environmental ratings are in validation.

GRADIENT-05 ROBOT

ENVELOPE AND SIDE VIEW

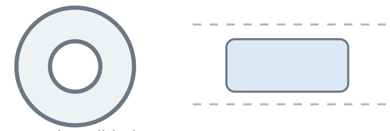
GradientOS kinematic envelope

tool0 origin reference



Horizontal reach approx. 1,509 mm

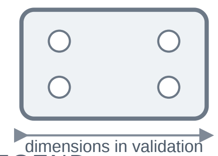
TOOL / FLANGE



tool interface data

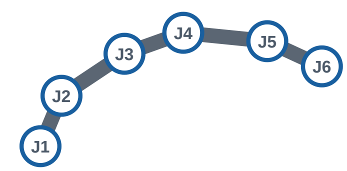
pattern in validation

BASE MOUNT



mounting interface in validation

AXIS LEGEND



J1 base
J2 shoulder
J3 elbow
J4-J6 wrist/tool

AXIS SPECIFICATIONS

Axis	Role	Motion Range	Rated Speed	Peak Speed	Ratio
J1	Base	+/- 361.0 deg	180 deg/s	360 deg/s	100:1
J2	Shoulder	+/- 108.9 deg	180 deg/s	360 deg/s	100:1
J3	Elbow	-240.6 / +87.7 deg	180 deg/s	360 deg/s	100:1
J4	Wrist roll	+/- 361.0 deg	1,000 deg/s	2,000 deg/s	18:1
J5	Wrist pitch	+/- 361.0 deg	1,980 deg/s	3,960 deg/s	100:11
J6	Tool roll	+/- 573.0 deg	1,800 deg/s	3,600 deg/s	10:1

GENERAL SPECIFICATIONS

Item	Unit	Gradient-05
Controlled axes	count	6
Horizontal reach	mm	approx. 1,509
Vertical reach span	mm	approx. 2,291
Encoder resolution	counts/rev	131,072 motor-side
Payload	kg	**In validation**
Repeatability	mm	**In validation**
Robot mass	kg	**In validation**
Power requirements	-	**In validation**
Controller	-	GradientOS / EtherCAT RTCore

RATINGS IN VALIDATION

- Payload rating
- Repeatability result
- Robot mass
- Wrist moment and inertia
- Tool flange interface
- Base mounting interface
- Power requirements
- Environmental rating
- Cable/air routing
- Production drawing package

DATA BASIS

Axis, limit, ratio, encoder, reach, and envelope data reflect the current GradientOS robot configuration and kinematic model.